Multi-Node Private Cloud Infrastructure Using OpenStack and Kolla-Ansible

Project Overview and Technical Implementation

Project period: March 2026 – September 2026

Abstract

Designed and expanded a multi-node private cloud environment using OpenStack Antelope (2023.1) and Kolla-Ansible on Ubuntu Server 22.04. The platform provides shared compute, networking and storage resources for student users, together with user-facing cloud services for file storage, monitoring and database access. The infrastructure began with Controller, Compute1 and Storage nodes and was expanded with a Dell Precision workstation acting as Compute2.

Project Objectives

- Build a multi-node OpenStack environment using dedicated physical nodes.
- Provide isolated cloud resources and services for multiple student users.
- Expand compute capacity by integrating an additional high-performance workstation.
- Expose selected services securely through a custom domain and Cloudflare Tunnel.
- Provide storage, monitoring, database and application-hosting capabilities.

Infrastructure Architecture

The core OpenStack environment is deployed across four physical nodes:

- Controller: Hosts OpenStack control-plane and API services.
- Compute1: Provides virtual machine compute capacity.
- Storage: Provides storage resources for the cloud environment.
- Compute2: An additional Dell Precision 7960 workstation integrated to increase available compute and storage capacity.

Compute2 Hardware Expansion

The additional Dell Precision 7960 workstation was provided by Samsung Innovation Lab for the cloud project. Its reported hardware configuration includes an Intel Xeon w9-3475X processor with 36 cores and 72 threads, 256 GB RAM and approximately 13 TB of raw storage capacity. The workstation was configured as the Compute2 node, extending the infrastructure beyond the original three-node deployment.

OpenStack Platform and Networking

OpenStack Antelope (2023.1) was deployed using Kolla-Ansible, with services running in Docker containers. The environment uses core OpenStack services for compute, networking, identity, image management and storage.

- Nova: Virtual machine provisioning and lifecycle management.
- Neutron: Tenant networking, virtual routers and network connectivity.
- Keystone: Identity, authentication and project-based access.
- Glance: Virtual machine image management.
- Cinder and Swift: Block storage and object storage.
- Skyline and Horizon: User-facing and administrative cloud dashboards.

Networking was implemented using Open vSwitch, VXLAN tenant networks, provider networking, routing and floating IP connectivity. The environment was configured to separate tenant workloads and provide controlled access to cloud resources.

Cloud Services for Student Users

The infrastructure was extended beyond basic virtual machine provisioning by making selected services available to classmates through dedicated web endpoints under openstacklab.me.

Cloud dashboard – cloud.openstacklab.me

Configured OpenStack Skyline as the student-facing dashboard. Horizon administrative access was kept restricted to the cloud providers.

Object storage – drive.openstacklab.me

Integrated OpenStack Swift with Nextcloud to provide a familiar file-management interface. Student users can access their object storage through the web interface and configured Android/iOS mobile applications, with user data separated by account and access permissions.

Monitoring – monitor.openstacklab.me

Implemented Prometheus and Grafana monitoring. User-facing dashboards are scoped to the relevant projects and virtual machines, with 24 panels selected for the student view. A more extensive administrative Grafana environment is maintained for cloud operators.

Database service – db.openstacklab.me

Configured relational database service access so users can obtain database environments for their projects.

Infrastructure as Code

Tested OpenStack Heat templates to provision infrastructure from declarative templates, similar in concept to cloud infrastructure templates offered by public cloud platforms.

Remote Access and Service Exposure

Configured Cloudflare Tunnel to publish selected cloud services through the custom domain openstacklab.me. Dedicated subdomains were configured for the Skyline dashboard, Nextcloud storage, Grafana monitoring and database service. The tunnel-based approach provides remote access without requiring each service to be exposed directly through a public inbound network route.

DevOps and Developer Services

- Integrated GitLab and Jenkins with Dokku: code pushed to GitLab triggers Jenkins pipeline steps, which then deploy the application to Dokku.
- Deployed Jupyter as a service with Keycloak-based authentication.
- Provided Coder as a browser-based development environment, deployed within the Kubernetes environment.

Kubernetes Workload Hosted on OpenStack

A separate four-VM Kubernetes cluster was provisioned on top of the OpenStack infrastructure. It consists of one control-plane VM and three worker VMs, each configured with 24 GB RAM and 6 vCPUs. Kubernetes was deployed using kubeadm, with containerd as the runtime and Calico for pod networking. This Kubernetes environment is a workload hosted by OpenStack, not an OpenStack service, and is documented separately as its own project.

Resilience and High-Availability Considerations

The platform was designed with logical resilience and service-continuity principles, including tenant separation, workload replication and recovery testing in the Kubernetes environment. A fully redundant OpenStack control plane was not implemented because the available hardware resources were prioritized for the deployed services and workloads. Accordingly, the environment should be described as a multi-node private cloud with HA-oriented design considerations, rather than a fully high-availability OpenStack control plane.

Implementation Challenges

The project involved troubleshooting across the cloud deployment, networking, storage and service-integration layers. Work included resolving Kolla-Ansible and Ansible dependency issues, configuring Open vSwitch and tenant networking, integrating additional compute hardware, exposing selected services through Cloudflare Tunnel, and configuring user-scoped access to shared services.

Results and Outcomes

- Expanded a three-node OpenStack environment into a four-node physical infrastructure by adding Compute2.
- Established a multi-tenant private cloud environment for student use.
- Provided a Skyline-based cloud dashboard and restricted administrative access.
- Made Swift-backed file storage accessible through Nextcloud web and mobile applications.
- Provided project- and VM-scoped monitoring dashboards through Grafana.
- Configured relational database access and tested Heat-based infrastructure provisioning.
- Integrated GitLab, Jenkins and Dokku for an application deployment workflow.
- Hosted additional developer services, including Jupyter, Keycloak and Coder.
- Provisioned a separate Kubernetes cluster on OpenStack virtual machines.

Technologies Used

Cloud: OpenStack Antelope (2023.1), Kolla-Ansible, Nova, Neutron, Keystone, Glance, Cinder, Swift, Skyline, Horizon, Heat.
Infrastructure: Ubuntu Server 22.04, Linux, Docker, Ansible, Dell Precision 7960, Open vSwitch, VXLAN.
Storage and Services: Nextcloud, Prometheus, Grafana, relational database services.
Platform and DevOps: Kubernetes, kubeadm, containerd, Calico, GitLab, Jenkins, Dokku, Jupyter, Keycloak, Coder.
Remote Access: Cloudflare Tunnel and custom DNS subdomains.

Conclusion

This project demonstrates the design, deployment and expansion of a multi-node private cloud using OpenStack. By integrating additional physical compute capacity and providing storage, monitoring, database and developer services, the environment was extended into a shared cloud platform for student workloads. The work provided hands-on experience in cloud infrastructure, Linux administration, software-defined networking, storage integration, multi-tenancy, service exposure and DevOps workflows.

Project Scope Note

The Kubernetes cluster is deployed as a separate workload on OpenStack-provisioned virtual machines and is intentionally treated as a separate project for portfolio presentation. High-availability wording in this report refers to design considerations and workload-level resilience, not a fully redundant OpenStack control plane.